

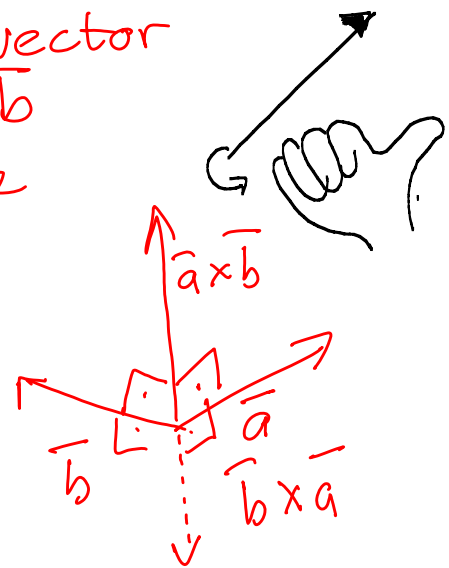
Cross product is the third type of multiplication

First: A 2×2 -determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad \text{eg } \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -2.$$

It can be pos, neg or zero.

The cross-product $\vec{a} \times \vec{b}$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$ with direction according to the right-hand rule.



Finding the cross-product:

Eg. $\langle 2, 1, 3 \rangle \times \langle 1, 2, 1 \rangle$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 3 \\ 1 & 2 & 1 \end{vmatrix} = -5\vec{i} - (-1)\vec{j} + 3\vec{k} \\ = \langle -5, 1, 3 \rangle.$$

General: If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$

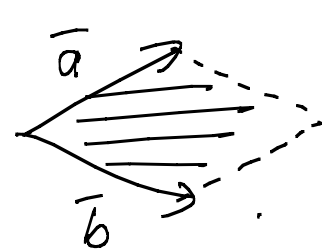
then $\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

$$= \langle a_2 b_3 - a_3 b_2, -a_1 b_3 + a_3 b_1, a_1 b_2 - a_2 b_1 \rangle$$

Theorems & properties

Let $\vec{a}$ & $\vec{b}$ be vectors in $\mathbb{R}^3$, $\vec{a} \neq \vec{0}$, $\vec{b} \neq \vec{0}$

1. $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

2. $\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0}$ 

3. $|\vec{a} \times \vec{b}|$ is the area of the parallelogram

4. $\vec{a} \times r\vec{b} = r(\vec{a} \times \vec{b}) = r\vec{a} \times \vec{b}$

5. $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$

NB The commutative property does not hold because $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$ but the distributive property does.

Examples : $\vec{a} = \langle 2, 1, 3 \rangle$, $\vec{b} = \langle 1, 2, 1 \rangle$

$$\vec{b} \times \vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ 2 & 1 & 3 \end{vmatrix} = \langle 5, -1, -3 \rangle$$

$$\vec{a} \times \vec{b} = \langle -5, 1, 3 \rangle$$

$$\vec{a} \times 2\vec{b} = 2\vec{a} \times \vec{b} = \langle -10, 2, 6 \rangle$$

$$\vec{a} \times \vec{a} = \vec{0}$$

Ex (i) Find a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$

$\vec{a} \times \vec{b} = \langle -5, 1, 3 \rangle$ is perpendicular to both $\vec{a}$ & $\vec{b}$.

$$|\vec{a} \times \vec{b}| = \sqrt{35}$$

$$\Rightarrow \text{Unit vector is } \left\langle \frac{-5}{\sqrt{35}}, \frac{1}{\sqrt{35}}, \frac{3}{\sqrt{35}} \right\rangle$$

$$\frac{\vec{a}}{|\vec{a}|}$$

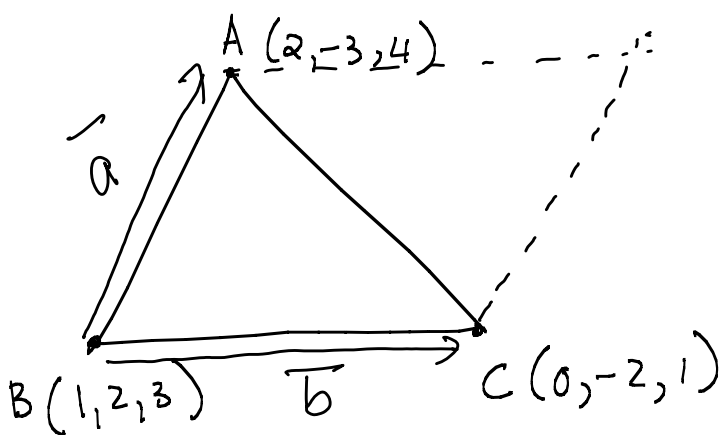
(ii) Find another unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

$$\left\langle \frac{5}{\sqrt{35}}, \frac{-1}{\sqrt{35}}, \frac{-3}{\sqrt{35}} \right\rangle$$

(iii) Find a vector of length 2 perpendicular to both $\vec{a}$ and $\vec{b}$.

$$\left\langle \frac{-10}{\sqrt{35}}, \frac{2}{\sqrt{35}}, \frac{6}{\sqrt{35}} \right\rangle$$

Ex Find the area of the triangle with vertices as given



$$\text{Area: } \frac{1}{2} |\vec{a} \times \vec{b}|$$

$$\vec{a} = \langle 1, -5, 1 \rangle, \quad \vec{b} = \langle -1, -4, -2 \rangle$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -5 & 1 \\ -1 & -4 & -2 \end{vmatrix} = \langle 14, 1, -9 \rangle$$

$$\frac{1}{2} |\vec{a} \times \vec{b}| = \frac{1}{2} \sqrt{278}$$

eg.

$$\begin{aligned} \vec{i} \times \vec{j} &= \vec{k} \\ \vec{k} \times \vec{i} &= \vec{j} \end{aligned}$$